

vasic-digital tmux — v1.0.20 Release Notes

Revision: 1 **Last modified:** 2026-05-29T16:30:00Z **Version:** 1.0.20 / versionCode 21 **Released:** 2026-05-29 **Hosts validated:** Mistborn (Darwin arm64) + nezha (ALT Linux 6.12 x86_64)

Summary

v1.0.20 completes the copy/paste story for tmx sessions. **Operator-confirmed this cycle:** drag-select text, then Cmd-V pastes the selected content. It fixes the `prefix P` paste binding (which previously produced "a completely new value"), and — the actual reason copy kept failing across earlier "fixes" — it makes config changes reach **already-running** sessions, so re-opening a long-lived session no longer leaves it stuck on stale bindings.

Together with v1.0.18 (plain-drag select+copy in mouse-tracking apps like Claude Code) and the `prefix m` mouse toggle, select + copy + paste now work end-to-end with no modifier gymnastics.

Fixed

`prefix P` paste — POSIX-clean rewrite

The previous binding:

```
bind P run -b 'tmux load-buffer - <<< "$({@clip-read})" \; tmux
paste-buffer -p'
```

was broken three ways:

1. **Nested-quote collision** — `{@clip-read}` expands to a *double-quoted* `sh -c "..."` command; substituting it inside the already double-quoted `"$({@clip-read})"` collided the quotes and mangled the captured text → the operator's "completely new value".
2. `<<<` **herestring** is a bash-ism; `run-shell` uses `/bin/sh`, which on Linux is `dash` → the command fails outright.
3. `\;` was passed literally to `sh`.

Rewritten collision-free and POSIX:

```
bind P run -b 'sh -c "pbpaste 2>/dev/null || wl-paste -n 2>/dev/null || xclip -o -selection clipboard 2>/dev/null || termux-clipboard-get 2>/dev/null" | tmux load-buffer - && tmux paste-buffer -p'
```

A single-quoted `run-shell` wraps a double-quoted inline OS-adaptive probe, piped to `tmux load-buffer -` then `paste-buffer -p` (bracketed). No `<<<`, no format nesting. Verified: real prefix `P` pastes the EXACT clipboard value, on macOS and Linux.

Stale-running-session config — the true root cause

A tmux server loads its config only at startup. A long-lived session re-opened via the shell-init prompt (`tmx attach -t NAME`) therefore kept its OLD mouse binding indefinitely — which is why "select/copy" kept failing for the operator even after the config was fixed (a brand-new session worked; the re-opened one did not). Two fixes:

- **tmx attach auto-reloads** the shipped config (source-file) into the session before attaching, so re-opening ANY existing session always applies the current bindings. Non-fatal/silent if absent.
 - **NEW tmx reload [-t NAME]** subcommand source-files the config into running sessions on demand (skips dead socket files via a has-session precheck), without killing the session or losing pane content.
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Tests — real-mouse automation with captured evidence

Per the operator mandate ("full automation tests which will really test this scenario and confirm it working with real evidence"):

- **test 58** (`58_operator_path_select_copy_ls.sh`) — drives the operator's EXACT scenario: real `tmx new -s Test` (wrapper / operator-path) → `ls` → a REAL SGR-1006 left mouse drag injected over the plain-shell output → asserts the dragged text reached the `@clip` sink; then proves `tmx attach` refreshes a deliberately-stale binding. 3/3 deterministic.
- **test 57** (`57_reload_select_copy_paste.sh`) — (A) stale-session reproduces the failure, (B) reload fixes copy, (C) paste-buffer mechanism, (D) REAL prefix `P` pastes the EXACT OS-clipboard value, (E) no-`<<<` POSIX guard.
- **test 56** — plain drag selects+copies inside a mouse-tracking app.
- `verify.sh` Layer-1 gate + test 46 T4 updated to accept the POSIX paste binding and FAIL on `<<<`; added gates for the plain-drag override + prefix `m`.
- Paired mutations: `M-PASTE`, `M-TMX-ATTACH-RELOAD`, `M-PLAINDRAG`, `M-MOUSETOGGLE`. Meta-test: 50 CAUGHT / 0 ESCAPED.

All headless (no GUI automation) — deterministic and CI-safe.

Validation (dual-host)

Host	install	suite	meta	copy/paste
Mistborn (Darwin arm64)	setup.sh GREEN	run_all GREEN	50 CAUGHT / 0 ESCAPED	operator- confirmed (pbpaste + Cmd- V)
nezha (ALT Linux x86_64)	setup.sh GREEN	run_all GREEN	0 ESCAPED	headless real- mouse proof PASS

Known pre-existing full-suite-load flakes (tests 16 window-name, 38 stale-pwd) remain tracked for a poll-with-budget hardening; both pass 3/3 standalone and are unrelated to the copy/paste fixes.

Files changed

- scripts/tmux.conf.template — POSIX prefix P paste binding.
- scripts/tmx.template — tmx reload subcommand + tmx attach auto-reload.
- scripts/tests/57_reload_select_copy_paste.sh (NEW),
58_operator_path_select_copy_ls.sh (NEW).
- scripts/tests/46_paste_in_physical.sh (T4), scripts/verify.sh
(Layer-1 gates).
- scripts/tests/meta_test_false_positive_proof.sh — M-PASTE + M-
TMX-ATTACH-RELOAD.
- VERSION, CHANGELOG.md, CONTINUATION.md, evidence under docs/qa/
2026-05-29-paste-fix-reload/.

Sources verified 2026-05-29

- tmux upstream man page — <https://man.openbsd.org/tmux.1>: run-shell
executes via /bin/sh; load-buffer - reads stdin into a buffer; paste-
buffer -p bracketed paste; set -g @user-option; source-file re-reads
config into a running server. <<< confirmed bash-only (not POSIX sh).